

AMAR SINGH

Senior Infrastructure Site Reliability Engineer

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PROFESSIONAL SUMMARY

Results-driven Staff SRE with 10+ years of expertise designing, implementing, and maintaining highly available distributed systems at scale. Proven track record of reducing operational overhead through automation, improving system reliability, and leading cross-functional teams to deliver mission-critical infrastructure projects. Deep expertise in cloud platforms (Openstack, GCP, AWS), container orchestration (Kubernetes, Rancher), and infrastructure-as-code (Terraform, Ansible).

PROFESSIONAL EXPERIENCE

Staff Infrastructure Site Reliability Engineer

Netskope | Santa Clara, CA | July 2019 – Present

- Led architecture review and production onboarding for 20+ critical microservices, collaborating with Principal Architects to ensure scalability, security, and operational readiness across GCP and OpenStack environments
- Designed and implemented disaster recovery strategy for critical production systems, reducing potential downtime from hours to minutes and ensuring business continuity
- Automated infrastructure provisioning and configuration management using Terraform and Ansible, reducing deployment time by 60% and eliminating configuration drift across 500+ instances
- Built comprehensive CI/CD pipelines with Jenkins and Git, enabling automated testing and deployment for infrastructure changes with 99.9% success rate
- Spearheaded adoption of Kubernetes, HashiCorp Vault, and Grafana monitoring, providing hands-on training to L1/L2/L3 operations teams and creating comprehensive documentation
- Performed root cause analysis on critical production incidents, implementing permanent fleet-wide fixes that reduced similar incidents by 85%
- Partnered with development teams to optimize application performance, identifying and resolving bottlenecks that improved response times by 40%
- Maintained 24x7 production support through on-call rotation, consistently meeting SLA targets with average incident response time under 5 minutes

Senior System Administrator

Nuvizz Inc. | Atlanta, GA | October 2017 – June 2019

- Architected and deployed production Kubernetes clusters across DEV, QA, and PROD environments, training operations staff on cluster administration and best practices
- Implemented container orchestration using Docker and Rancher, streamlining application deployment and reducing environment provisioning time from days to hours
- Deployed and maintained high-availability database clusters (MySQL, Couchbase, Elasticsearch, RabbitMQ) ensuring 99.95% uptime
- Built centralized monitoring infrastructure using Zabbix, creating custom alerts for 200+ services, systems, and network devices
- Automated backup and recovery operations using Bacula for 100+ Windows and Linux servers, with regular testing to ensure data integrity
- Established CI/CD workflows with Jenkins and GitLab, enabling automated testing and deployment for infrastructure changes

System Administrator

Hewlett-Packard Inc. | Palo Alto, CA | November 2016 – October 2017

- Managed VMware ESXi infrastructure and SAN storage, provisioning virtual environments for development and production workloads
- Administered AWS EC2 instances and cloud resources, optimizing costs while maintaining performance requirements
- Configured and maintained HP ProLiant blade servers and HP ProCurve 5580 switches for corporate and data center networks
- Provided enterprise-level technical support for server infrastructure, resolving complex issues and minimizing downtime

TECHNICAL SKILLS

Cloud & Infrastructure: GCP, AWS, OpenStack, VMware ESXi, KVM

Container & Orchestration: Kubernetes, Docker, Rancher, Container Security

Automation & IaC: Terraform, Ansible, Jenkins, CI/CD Pipelines, Git/GitLab

Monitoring & Observability: Grafana, Prometheus, Zabbix, Nagios

Databases: MySQL, Couchbase, Elasticsearch, RabbitMQ

Web Services: Apache, Nginx, HAProxy, Load Balancing

Operating Systems: Linux (Ubuntu, CentOS, RHEL), Windows Server

Scripting & Programming: Python, Bash, Shell Scripting

Networking: TCP/IP, DNS, DHCP, HTTPS, SSL/TLS, VLANs, STP, VTP

Security: HashiCorp Vault, Security Hardening, Compliance

EDUCATION

Master of Science in Network Communication

University of Houston | Houston, TX | 2014 – 2016

ACADEMIC PROJECTS

SDN Traceroute: Tracing SDN Forwarding without Changing the Network Behavior(08/2014 - 05/2015)

- SDN traceroute tha. can query the current path taken by any packet through an SDN-enabled network.
- The path is traced by using the actual forwarding mechanism at each SDN-enabled device without changing the forwarding rules being measured.
- This enables administrators to discover the forwarding behavior for arbitrary Ethernet packets, as well as debug problems in both switch and controller logic.

HTCondor: High Throughput Cluster Computing (08/2015 - 05/2016)

- HTCondor is a software system that creates a High-Throughput Computing (HTC) environment.
- It effectively uses the computing power of machines connected over a network, be they a single cluster, a set of clusters on a campus, cloud resources either standalone or temporarily joined to a local cluster, or international grids.

CERTIFICATIONS

• Cisco Certified Network Professional (CCNP) – Routing • Cisco Certified Network Associate (CCNA) • Python Programming (Rice University)